

Mahalanobis distance notes

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1 Mahalanobis Distance as a Statistical Decision Rule

This section explains how Mahalanobis distance can be converted into a statistically principled decision rule using the Chi-square distribution.

1.1 Core Idea

For a d -dimensional multivariate normal distribution, the squared Mahalanobis distance is defined as:

$$D^2 = (x - \mu)^T \Sigma^{-1} (x - \mu)$$

Under the assumption of multivariate normality, this quantity follows a Chi-square distribution:

$$D^2 \sim \chi^2(d)$$

Therefore, instead of choosing an arbitrary distance threshold, a statistically valid threshold can be obtained from the Chi-square distribution.

1.2 Mahalanobis Threshold Using the Chi-square Distribution

1.2.1 Step 1: Dimensionality

Let:

- d denote the number of features.

Examples:

- 2 features $\Rightarrow d = 2$
- 10 features $\Rightarrow d = 10$

1.2.2 Step 2: Significance Level

A significance level α is chosen to control the strictness of the decision rule.

α	Interpretation
0.10	Loose (90% of data inside)
0.05	Standard (95% of data inside)
0.01	Strict (99% of data inside)

1.2.3 Step 3: Chi-square Critical Value

The threshold is defined as the $(1 - \alpha)$ quantile of the Chi-square distribution:

$$\chi_{d, 1-\alpha}^2$$

Example critical values are shown below:

d	$\alpha = 0.05$	$\alpha = 0.01$
2	5.99	9.21
5	11.07	15.09
10	18.31	23.21

1.2.4 Step 4: Decision Rule

An observation is classified as follows:

$$D^2 \leq \chi_{d, 1-\alpha}^2 \Rightarrow \textit{In - distribution}$$

$$D^2 > \chi_{d, 1-\alpha}^2 \Rightarrow \textit{Outlier/PrivacyRisk}$$

1.3 Concrete Numeric Example (2D)

Consider:

- Number of features: $d = 2$
- Significance level: $\alpha = 0.05$
- Threshold: $\chi_{2, 0.95}^2 = 5.99$

Sample	D^2	Verdict
A	2.3	Normal
B	4.8	Normal
C	6.4	Outlier
D	12.1	High Risk

1.4 Implications for Privacy and Synthetic Data

In the context of synthetic data evaluation:

- If the Mahalanobis distance between a synthetic sample and the real data distribution satisfies:

$$D_{syn \rightarrow real}^2 < \chi_{d,0.95}^2,$$

- then the synthetic record may be statistically indistinguishable from real data,
- indicating a potential membership inference risk.

The proportion of synthetic samples that fall inside the Chi-square confidence ellipsoid can be used as:

- a privacy leakage score,
- an overlap risk metric.

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1.5 Practical Computation

The Chi-square threshold can be computed programmatically as:

$$\chi_{d,1-\alpha}^2 = \text{Quantile}_{1-\alpha}(\chi_d^2)$$

For example, using Python:

```
from scipy.stats import chi2

d = 10
alpha = 0.05
threshold = chi2.ppf(1 - alpha, df=d)
print(threshold)
```

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1.6 Key Intuition

Mahalanobis distance squared follows a Chi-square distribution; therefore, Chi-square critical values provide statistically valid thresholds for outlier detection and privacy risk assessment.

2 Mahalanobis Distance Threshold Using the Chi-square Distribution

2.1 Chi-square Threshold Definition

To determine whether a point is statistically consistent with the reference distribution, a threshold is chosen based on a significance level α :

$$D^2 \leq \chi_{d, 1-\alpha}^2$$

Points exceeding this threshold are considered statistically unlikely (outliers).

2.2 Example: Two-dimensional Case

Let:

- Number of features: $d = 2$
- Significance level: $\alpha = 0.05$

The Chi-square threshold is:

$$\chi_{2,0.95}^2 = 5.99$$

This value satisfies:

$$P(\chi_2^2 \leq 5.99) = 0.95$$

2.3 Derivation of $\chi_{2,0.95}^2 = 5.99$

The cumulative distribution function (CDF) of a Chi-square distribution with $k = 2$ degrees of freedom is:

$$F(x; 2) = 1 - e^{-x/2}$$

To find the 95th percentile, solve:

$$1 - e^{-x/2} = 0.95$$

$$e^{-x/2} = 0.05$$

$$-\frac{x}{2} = \ln(0.05)$$

$$x = -2 \ln(0.05) \approx 5.99$$

Thus:

$$\chi_{2,0.95}^2 = 5.99$$

2.4 Decision Rule

For a given observation:

$$\{ D^2 \leq 5.99 \text{ In-distribution(normal)} D^2 > 5.99 \text{ Outlier/statistically unlikely} \}$$

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2.5 Interpretation in Privacy and Synthetic Data

In the context of synthetic data privacy:

- If a synthetic sample has D^2 below the Chi-square threshold relative to real data,
- it may be statistically indistinguishable from real records,
- indicating potential membership inference risk.

Counting the proportion of synthetic samples inside the Chi-square ellipsoid provides a quantitative privacy risk indicator.

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2.6 Practical Computation

The Chi-square threshold can be computed programmatically as:

$$\chi_{d,1-\alpha}^2 = \text{Quantile}_{1-\alpha}(\chi_d^2)$$

For example, in Python:

```
from scipy.stats import chi2
chi2.ppf(0.95, df=2)
```